

Conversion formulas included in this spreadsheet:

Diameter to AWG	AWG()
AWG to diameter	DIAMETER()
Thickness to copper plating weight	CPW()
Copper plating weight to thickness	THICKNESS()

Resistance formulas included in this spreadsheet:

DC resistance of round wires	
From diameter	RROUND()
From AWG wire size	RROUND_AWG()
At room temperature only	RROUND_RT()
DC resistance of printed circuit board traces	
From trace thickness and width	RTRACE()
Using copper plating weight	RTRACE_CPW()
At room temperature only	RTRACE_RT()
DC resistance of power or ground planes	
Using thickness and via diameter	RPLANE()
Using copper plating weight	RPLANE_CPW()

Variables used:

ρ Bulk resistivity of copper $\rho := 6.787 \cdot 10^{-7}$ ohm-in.

This coefficient is slightly different from the bulk resistivity of pure copper ($6.58E-07$) owing to the annealing process used in making wire, and chemical imperfections in the copper used for making practical wires.

In practice, the resistance of two wires making up a twisted pair may often be matched as well as 10%, but almost never as well as 1%.

$\delta\rho$ Thermal coefficient of resistance $\delta\rho := .0039$ per deg. C

If the resistance of a copper wire is R at room temperature, then at a temperature 1°C higher it will be $R(1 + \delta\rho)$. This coefficient applies to standard annealed copper wires. The coefficient for pure copper in its bulk state varies slightly.

Over a temperature range $0-70^\circ\text{C}$ the resistance of copper wires varies 28%.

x Length of wire (in.)
(or separation between contact points on ground plane)

d Diameter of wire (in.)
(or diameter of contact point on ground plane)

AWG American wire gauge (English units)

temp Temperature (°C)

w Width of printed circuit board trace (in.)

t Thickness of printed circuit board trace (in.)

cpw Thickness of printed circuit board traces, in units
of copper plating weight (oz/ft²)

Conversions between American
Wire Gauge (AWG) and diameter (in.): $\text{DIAMETER}(\text{awg}) := 10^{-\frac{\text{awg}+10}{20}}$

$$\text{AWG}(d) := -10 - 20 \cdot \log(d)$$

General formula for
resistance of a round wire (Ω):

$$\text{RROUND}(d, x, \text{temp}) := \frac{4 \cdot \rho \cdot x}{\pi \cdot d^2} \cdot [1 + (\text{temp} - 20) \cdot \delta \rho]$$

Resistance of a round wire specified
by AWG size instead of diameter (Ω):

$$\text{RROUND_AWG}(\text{awg}, x, \text{temp}) := \text{RROUND}(\text{DIAMETER}(\text{awg}), x, \text{temp})$$

Resistance of a round wire
at room temperature (Ω):

$$\text{RROUND_RT}(d, x) := \text{RROUND}(d, x, 20)$$

Conversion between thickness,
t (in.) and copper plating
weight, cpw (oz):

$$\text{CPW}(t) := \frac{t}{.00137}$$

$$\text{THICKNESS}(\text{cpw}) := .00137 \cdot \text{cpw}$$

Resistance of a
circuit trace (Ω):

$$\text{RTRACE}(w, t, x, \text{temp}) := \frac{x \cdot \rho}{w \cdot t} \cdot [1 + (\text{temp} - 20) \cdot \delta \rho]$$

Resistance of a trace specified
by plating weight instead of thickness (Ω):

$$\text{RTRACE_CPW}(w, \text{cpw}, x, \text{temp}) := \text{RTRACE}(w, \text{THICKNESS}(\text{cpw}), x, \text{temp})$$

Resistance of a circuit trace
at room temperature (Ω):

$$\text{RTRACE_RT}(w, t, x) := \text{RTRACE}(w, t, x, 20)$$

Resistance of a power or ground plane (Ω):

When using long, skinny traces or wires, the approximations above work extremely well. Each formula assumes a uniform distribution of current throughout the conducting body, for which resistance is directly proportional to length.

Currents circulating in a large ground or power plane are not uniform. Consequently, the resistance measured between two points on a ground or power plane is not directly proportional to the separation between measurement points.

The following equation models the resistance between two contact points on a ground plane. This model assumes each contact point touches the ground plane over some finite area. The approximate diameter of the contact point determines the overall resistance.

If the contact points lie near any edge of the plane, the resistance between them may go up by a factor of 2. The resistance near corners may rise even higher.

d1 Diameter of 1st contact point (in.)
d2 Diameter of 2nd contact point (in.)
t Thickness of plane (in.)
cpw Thickness of plane, copper plating weight (oz)
x Separation between contact points (in.)
temp Temperature ($^{\circ}\text{C}$)

Resistance of a power or ground plane (Ω):

$$\text{RPLANE}(d1, d2, t, x, \text{temp}) := \frac{\rho}{2 \cdot \pi \cdot t} \cdot \left(\ln\left(\frac{2 \cdot x}{d1}\right) + \ln\left(\frac{2 \cdot x}{d2}\right) \right) \cdot [1 + (\text{temp} - 20) \cdot \delta\rho]$$

Resistance of a power or ground plane specified
by plating weight instead of thickness (Ω):

$$\text{RPLANE_CPW}(d1, d2, \text{cpw}, x, \text{temp}) := \text{RPLANE}(d1, d2, \text{THICKNESS}(\text{cpw}), x, \text{temp})$$